

Recycling Image Classification: Case Study Rubric

DS 4002, Case Study CS3

Submission format: Upload link to GitHub repository on UVA Canvas

Why am I doing this?

This case study gives you hands-on experience building and evaluating a deep learning image classification pipeline on real-world data. You will work with the RealWaste dataset (actual landfill photographs) and apply convolutional neural network (CNN) techniques to solve a practical environmental problem. Along the way you will practice key ML skills: data preprocessing, model design, transfer learning, and performance evaluation.

What am I going to do?

The GitHub repository for this case study contains the starter code, data dictionary, and reference materials. You will preprocess the RealWaste images, map the nine original material categories into a binary recyclable/non-recyclable label, and train at least two CNN-based models. Your final deliverables include:

- Well-documented, commented Python source code (Jupyter Notebook or .py scripts)
- A written report PDF summarizing your approach, results, and conclusions
- A GitHub repository containing all materials
- A REFERENCES.md file in IEEE citation style

How will I know I have succeeded?

You will meet expectations on this case study when you follow all criteria below.

Formatting	One GitHub repository submitted via link on Canvas. Repository must include: • README.md • LICENSE.md • Folder with source code • Data folder (or link) • REFERENCES.md • Written report PDF
README.md	Brief summary of the project, dataset, and your approach. Should orient a new reader to the repository in a few sentences. Include a link to the dataset.
Source Code File	Well documented Notebook file that contains the code used to execute your image and statistical analyses. In the source code, you must include: - Data preprocessing (including resizing, data augmentation,

	and train/test split) - Exploratory data analysis (bar charts and sample images) - Modeling baseline CNN (Must include at least 2 convolutional layers and appropriate pooling. Report training and validation accuracy/loss curves. Evaluate on the test set.) - Modeling transfer learning (Apply transfer learning using a pretrained architecture (MobileNetV2 or ResNet50 with ImageNet weights). Fine-tune as appropriate. Report training and validation accuracy/loss curves. Evaluate on the test set.)
Evaluation and Comparison	For both models report on the test set: Accuracy, Precision, Recall, and F1-score - Confusion matrix (Compare the two models and discuss which performed better and why) - Discuss what types of errors (false recyclable vs.false non-recyclable) are more costly
Written Report	A PDF (~3/4 pages, excluding figures and references) containing: - Problem summary and motivation - Description of your data preprocessing choices - Explanation of your modeling approach with a simple diagram - Results and comparison of both models - Conclusions and recommendations for the waste management company - Reflection: challenges faced and what you would do differently
REFERENCES.md	Markdown file citing all resources used in IEEE format. Include a brief annotation under each citation explaining how it helped you. Must include at least 3 references beyond those provided in the starter materials.